

Name: _____





BOOTSTRAP

Equity • Scale • Rigor

Brought to you by the Bootstrap team:

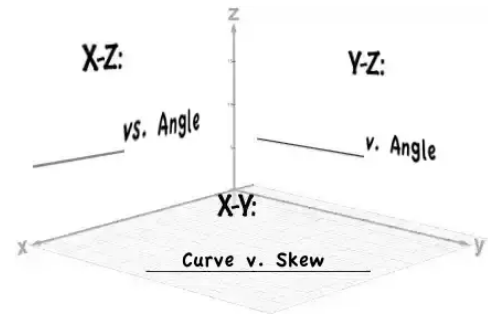
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Predicting Steering Angle from Curve *and* Skew

Make sure you have [Curve v. Skew v. Steering Angle \(Desmos\)](#) open in Desmos. Click on the wrench button (🔧) in the top-right corner to **Open "Graph Settings"**. At the bottom of the "More Options" list and check the box for **Translucent Surfaces**.

Working in 3 Dimensions

1) What do you Notice? What do you Wonder?



Click on the cube to drag and rotate it so that the z-axis points up, y-axis points to the right, and x-axis points towards you.

2) Now the 3D graph shows a (purple) 2D scatter plot. Which plane is the scatter plot on? x-y x-z y-z

3) Which variables are being compared in this plane? _____

4) Adjust m_{skew} and b to find the best fit. m_{skew} : _____ b : _____. As skew increases, the steering-angle _____

Rotate the cube to the right, so that the z-axis points up, y-axis points to the right, and x-axis points towards you.

5) What did the line look like while you were rotating the cube? _____

6) Why do the purple dots look like they're in one line? _____

Turn off the "Steering-Angle v. Skew (purple)" folder and turn on "Steering-Angle v. Curve (blue)"

7) Now the 3D graph shows a (blue) 2D scatter plot. Which plane is the scatter plot on? x-y x-z y-z

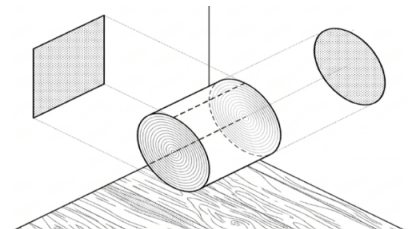
8) If you haven't already, label the planes on the 3D plot at the top of the page with the correct variable names.

9) Use the sliders to fit a line to the points in this plane. (Note: m_{curve} is beneath the curve folder!)

m_{curve} : _____ b : _____. As curve increases, the steering-angle _____

Making Connections

10) Set the sliders to zero, then "turn on" *all the folders*. Rotate the cube and explore. What do the orange dots have to do with the diagram on the right?



Fitting a 3D Model

11) "Turn off" the 2d projections (blue and purple) and focus on the orange dots. Set the slope sliders to the values you found for your the 2D plots and adjust the y-intercept slider to the **plane of best fit** for this data. Readjust the slope sliders if need be. Then use your slider estimates to complete the Pyret definition below:

```
fun steering-angle(r): ( _____ * r["curve"]) + ( _____ * r["skew"]) + _____ end
                        m_curve                      m_skew                      b
```

Supervised Learning - Multiple Regression

To complete this page, open your copy of [Self-Driving Car Starter File](#) and click "Run". You'll be using the following functions:

```
# regression-model-code :: ( Table, List<String>, String ) -> Code
                        |table-name|      |explanatory|    |response|
```

```
# regression-model-S :: ( Table, List<String>, String, Row->Number) -> Number
```

Building a Two-Variable Model

1) Evaluate `regression-model-code(training, [list: "curve", "skew"], "steering-angle")` to produce the predictor function, then fill in the model below. (Round your numbers to 2-3 digits)

```
fun curve-skew-predictor(r):
```

$$\left(\frac{\quad}{m_{\text{curve}}} * r[\text{"curve"}] \right) + \left(\frac{\quad}{m_{\text{skew}}} * r[\text{"skew"}] \right) + \frac{\quad}{b}$$

end

2) **Copy-and-paste** your model into the Definitions Area.

3) Use `regression-model-S` to compare the fit of your 1-variable and 2-variable models:

Model	S -value
<pre> regression-model-S(training, [list: "curve"], "steering-angle", curve-predictor) </pre>	
<pre> regression-model-S(training, [list: "curve", "skew"], "steering-angle", curve-skew-predictor) </pre>	

4) Adding **skew** to the model made the S-value larger/smaller so we know that the fit is better/worse.

Thinking Critically About Variables

5) If you had to **drop one variable** from the two-variable model, which would you drop, and why? _____

6) What do you think would happen to the S-value if you added all four variables — `curve`, `skew`, `offset`, and `speed` — to the model?

7) Try it out! What was the result? *S-value*: _____

Reflecting on Self-Driving Cars

Thinking about Training

1) Refer to the weather forecast (right) in your response. During a week of daytime test drives:

- On Wednesday and Friday, the self-driving car drives safely.
- On the other days, the self-driving car is unsafe.

What might explain why the car drove unsafely on some days?

SAT	SUN	MON	TUE	WED	THU	FRI
						
THUNDER, HEAVY RAIN	SUNNY, LIGHT SNOW	CLOUDY, LIGHT RAIN	HEAVY CLOUD	FEW CLOUDS	CLOUDY	SUNNY, WARMER
63°F 41°F	65°F 42°F	64°F 43°F	69°F 40°F	71°F 42°F	72°F 39°F	70°F 65°F

2) Imagine that a self-driving car has done extensive training on a one-lane road, in all weather conditions, and at all times of day. Would you expect it to be able to safely drive on a busy two-lane road? Explain.

3) Imagine that a self-driving car trained on isolated country roads, as well as city streets, and highways in all weather conditions and times of day. It would be pretty safe on many roads! What might it still not be prepared to navigate safely?

Confidence Rating

4) In addition to producing a steering angle for each image of the road, many models produces a *numeric* measure of "confidence" in its response.

- What do you think causes a model's "confidence" to increase or decrease?

- Is 100% "confidence" possible?

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